

International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE, Final Term Examination, Spring 2026
Course Code: CSE-2321 Course Title: Data Structures
Total Marks: 50 Time: 2 hours 30 minutes

*Answer all questions. Figures in the right-hand margin indicate full marks.
Use separate answer scripts for Group A and Group B.*

M CO DL

Group – A

1. a) Suppose a stack of integers is allocated a maximum of $M = 6$ memory cells in an array layout, and it is initially empty ($TOP = 0$). Trace the stack contents step-by-step and state the final value of TOP after executing the following sequential operations:

1. PUSH(STACK, 30)
2. PUSH(STACK, 60)
3. POP(STACK, VAL_1)
4. PUSH(STACK, VAL_1 + Z) (Note: Z is the last single digit of your student ID)
5. PUSH(STACK, 90)
6. POP(STACK, VAL_2)

3 CO1 C3

- b) Consider the following *infix* expression Q:

$$Q: ((M + N) * (O - P)) ^ (Q / R)$$

Convert the *infix* expression Q into its equivalent *postfix* expression P using the algorithm you studied.

3 CO2 C2

- c) Write an algorithm or program that uses a Stack (or its underlying push/pop logic) to check whether a string containing the bracket types {}, [], and {} is correctly nested and balanced. Print "Valid" if the sequence is balanced, and "Invalid" otherwise.

Input	Output
{{(())}}	Valid
{[]}	Invalid
{[(())]}	Valid

4 CO3 C4

OR

- c) An editor tracks document changes using an Undo Stack and a Redo Stack.

Use integer codes to represent operations: 1 for "Type" (followed by an integer value), 2 for "Undo", and 3 for "Redo". When a user performs Type, the value is pushed onto the Undo Stack and the Redo Stack is cleared. Undo moves the top value from the Undo Stack to the Redo Stack, while Redo moves the top value from the Redo Stack back to the Undo Stack.

Write a program to process a sequence of N operations and display the final contents of the Undo Stack.

Input	Output
4 1 10 1 20 2 3	10 20
4 1 5 1 15 2 1 25	5 25

4 CO3 C4

Answer all questions. Figures in the right-hand margin indicate full marks.
Use separate answer scripts for Group A and Group B.

M CO DL

2 a) Suppose a queue is maintained by a circular array **QUEUE** with 12 memory cells.

I. State the general formula to calculate the number of elements currently in the queue, assuming 1-based indexing.

II. Find the total number of elements currently in the **QUEUE** for the following conditions:

1+1 CO1 C2

- FRONT = 4, REAR = 9
- FRONT = 10, REAR = 2

b) What is a *circular queue*? Write the procedures for *inserting* an element into a circular queue and *deleting* an element from a circular queue. What are the *time complexities* of these operations?

3 CO1 C2

OR

b) Compare the Big-O time complexity for the **INSERT** and **DELETE** (extract-max/min) operations of a **Priority Queue** based on the following three implementation strategies: **Sorted Linked List**, **2D Array (Unsorted)**, and **Binary Heap**, and present your findings in a structured table.

3 CO1 C2

c) Write a recursive algorithm to compute the sum of all elements of an array. Trace the algorithm using the array {4, 7, 2, 9, 5}

1+2 CO2 C3

OR

c) Consider a recursive function tracking the cycle steps of the $3n + 1$ problem:

```
int collatz_steps(int n) {
    if (n == 1) return 1;
    if (n % 2 == 0) return 1 + collatz_steps(n / 2);
    else return 1 + collatz_steps(3 * n + 1);
}
```

3 CO2 C3

Manually trace the step-by-step execution states and calculate the exact return value of **collatz_steps(7)**.

d) Implement the **enqueue()** function of an array-based queue, and state its time complexity.

2 CO2 C3

Group – B

a) You are given the input array [170, 45, 75, 90, 802, 24]. Illustrate a step-by-step trace of **Radix Sort** (using base 10). Show the state of the buckets after each of the passes.

3 CO2 C3

b) Analyze and identify the most optimal sorting algorithm for each of the following scenarios, providing a concise justification based on performance efficiency:

3 CO3 C4

Answer all questions. Figures in the right-hand margin indicate full marks.
Use separate answer scripts for Group A and Group B.

M CO DL

- i. Processing a stream of nearly-sorted data where elements arrive sequentially with minimal inversions.
- ii. Sorting a large integer dataset restricted to a very small range (e.g., ages 1–100).
- iii. Sorting a massive dataset in a memory-constrained environment where a guaranteed $O(n \log n)$ worst-case time complexity is required without using extra space.

c) Given a sorted array with duplicate elements [2, 4, 7, 7, 7, 12, 15] (0-indexed), explain how the standard binary search algorithm must be modified to find both the **Lower Bound** and **Upper Bound** of a target value. State the specific indices returned by these two variants when searching for the target value 7 in the provided array.

4 CO2 C3

OR

c) A one-dimensional array is said to have a **peak element** if the element is greater than or equal to its adjacent elements. Explain how a modified **Binary Search** can be used to find a peak element in $O(\log n)$ time. Trace the algorithm on the array: [3, 7, 15, 18, 12, 8, 5] and identify the peak element.

4 CO2 C3

4. a) What is a *binary search tree (BST)*? Insert the following keys into an initially empty binary search tree. 55, 35, 70, 25, 40, 65, 80, 60, 38. Then delete 35 using the BST deletion algorithm and show the final tree.

3 CO2 C2

OR

a) The preorder and inorder traversal sequences of a binary tree T are given below:

Preorder: A, B, D, E, C, F, G, H

Inorder: D, B, E, A, F, C, G, H

Construct (draw) the binary tree T from the given traversals, showing your reasoning.

3 CO2 C2

b) Explain why the average time complexity for searching in a BST is $O(\log n)$, whereas the worst-case complexity is $O(n)$. Under what specific condition does the worst-case scenario occur?

2 CO1 C2

5. Consider the following array representation of a Max-Heap: [90, 70, 80, 40, 60, 10, 50]. Draw the tree representation of this heap.

3 CO1 C2

6. Which of the following data structures is most suitable for efficiently finding the top five largest salaries from a dataset containing five million employees? Justify your answer.

3 CO3 C4

- i) Max heap ii) BST iii) Min heap iv) AVL tree.

Answer all questions. Figures in the right-hand margin indicate full marks.
Use separate answer scripts for Group A and Group B.

M CO DL

5. a) Define the "Balance Factor" of a node in an AVL tree. If an insertion causes a node's balance factor to become +2 or -2, explain why a rotation is necessary to restore the height-balanced property.

3 CO1 C2

OR

- a) Describe the structural changes that occur during a Left-Right (LR) rotation and a Right-Left (RL) rotation in an AVL tree. Provide an example sequence of three keys that would trigger an RL rotation.

3 CO1 C2

- b) Insert the following keys into an initially empty AVL Search Tree: [20, 10, 40, 30, 50, 35]. Show every rotation performed, and draw the resulting tree.

2 CO1 C3

- c) Consider a hash table of size $M = 10$ intended to store employee IDs: [1122, 2122, 3122, 4122, 5122].

- Insert these keys into the table using the Division Method ($H(k) = k \bmod 10$) combined with Linear Probing. Show the final state of the table.
- Analyze the performance of this insertion.
- Propose an alternative (either a different hash function like Folding or a different collision resolution method like Quadratic Probing) and justify why it would produce a more uniform distribution for this specific dataset.

2+1+2 CO3 C4



International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
Semester Final Examination

Program: B. Sc. in CSE
Course Code: CSE 2323
Time: 2 hour 30 minutes

Semester: Spring -2026
Course Title: Digital Logic Design
Total Marks: 50

Answer the following questions. The figures in the right-hand margin indicate full marks.

Group-A

- | | | COs | Marks |
|---|---|------------|--------------|
| 1 | a) Design a 3-bit magnitude comparator that outputs $A > B$, $A = B$, and $A < B$. Derive the logic expressions and draw the circuit. | CO2 | 5 |
| | OR | | |
| | Design a 5-to-32 decoder using two 4-to-16 decoders: use the MSB to enable exactly one 4-to-16, and feed the remaining 4 bits to both decoders. | CO2 | 5 |
| | b) Design a multiplexer using the following Boolean function,
$F(A, B, C, D) = \sum(1, 3, 4, 6, 9, 11, 12, 13, 14)$ | CO2 | 5 |
| | OR | | |
| | Explain how an 8:1 MUX can be constructed using 4:1 MUX. Illustrate the design with a neat diagram. If $S_2S_1S_0 = 101$ is selected in the selection lines, then which input line will be selected in your design? | CO2 | 5 |
| 2 | a) How does a JK flip-flop eliminate the indeterminate state of an S-R flip-flop? Design a JK flip-flop and show its characteristic equation, characteristic table, logic diagram and timing diagram. | CO3 | 5 |
| | b) Describe the working of a master-slave D flip-flop. Include its logic diagram and characteristic table in your explanation. | CO3 | 5 |

Group-B

- 3 a) Compare the Ring Counter and Johnson Counter. Design a 4-bit asynchronous decade (MOD-10) counter using T flip-flops. CO3 5
- OR
- ① Design a 3-bit up-down counter using T flip-flops with control input μ ($\mu=1$ for up, $\mu=0$ for down). Draw the logic circuit with derived T flip-flops in equation. CO3 5
- b) A combinational circuit is defined by the following functions:
- $$F_1(A, B, C) = \sum(3, 5, 7)$$
- $$F_2(A, B, C) = \sum(4, 5, 7)$$
- CO3 5
- Implement the circuit with a PLA having 3 inputs, 3 product terms, and 2 outputs
- 4 a) Design a 4-bit Serial-In Parallel-Out (SIPO) shift register using D flip-flops. Explain its operation and draw the corresponding timing diagram. CO3 5
- b) Design a synchronous counter using T flip-flops to follow the sequence: 7, 4, 3, 1, 5, 0, 7, 4, 3, ... CO3 5
- Mention the design steps clearly, then solve step by step.
- 5 a) Write short notes on:
- Propagation delay CO2 3
 - ROM and its types
 - Race around condition.
- b) Write down the differences between SRAM and DRAM. CO3 2
- c) Design a 4X5 RAM and describe its operation. Analyze how many address lines are required to uniquely select a memory location in a 4X5 RAM and justify your answer. CO3 5
- OR
- Draw the circuit diagram, block diagram of Binary Storage Cell (BC). Explain how read and write operations are performed using the Select and Read/Write control signals. CO3 5

International Islamic University Chittagong
Department of Computer Science & Engineering

B.Sc. in CSE Final Examination, Spring - 2026

Course Title: Mathematics-III Course Code: MATH-2307 (New)

Course Title: Mathematics-IV Course Code: MATH-2401 (Old)

Time: 2 Hours 30 Minutes

Full Marks: 50

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Course Outcomes (COs) of the Questions	
CO1	Understand the fundamentals of Matrix, Linear system of equations & Vector analysis
CO2	Implement the fundamental knowledge of Matrix, linear system of equations, vector functions, vector field, scalar field, gradient, divergence, curl, differentiation and integration of vector valued functions, partial derivatives in different problems
CO3	Solve line integrals, surface area, surface integrals, volume integrals, and the work done in different problems
CO4	Apply Green's theorem, Stoke's theorem, Gauss' theorem in solving mathematical problems

Bloom's Levels of the Questions						
Letter Symbols	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Answer the following questions

Part A

1. Write the names of 4 methods of matrix decomposition. Verify whether the matrix can be Eigen Decomposed $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.

CLOs DL Marks
CLO1 An 2+8

Or

Solve the following set of linear equations using LU decomposition.

CLO1 An 10

$$x + 2y + 3z = 5, 2x - 4y + 6z = 18, 3x - 9y - 3z = 6$$

2. a) As for an example $\vec{OP} = 2\hat{i} + 4\hat{j} + 6\hat{k}$ makes an angle α, β and γ with the x-axis, y-axis and z-axis respectively, Find the angle α, β and γ

CLO2 U 5

- b) Find all vectors $\vec{v}$ such that $(\hat{i} + 2\hat{j} + \hat{k}) \times \vec{v} = 3\hat{i} + \hat{j} - 5\hat{k}$

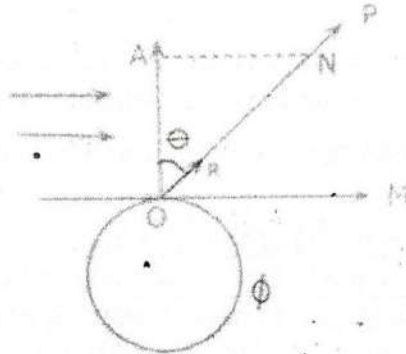
CLO2 U 5

Part B

CLOs DL Marks

CLO2 U 3

3 a)



From the graph above, write down which vector is the $\vec{\nabla} \phi$ vector

and which vector is the $d\mathbf{r}$ vector. If you want to find the directional derivative from the graph above, what should you do?

The directional derivative is equal to = ? Where $\mathbf{r}$ is a position vector, $\hat{\mathbf{R}}$ is a unit vector of $\vec{OP}$ and ϕ is a surface.

b) Find the values of a so that the vector $\vec{F} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x + az)\hat{k}$ is solenoidal.

CLO2 U 3

c) Find the values of a, b and c so that $\vec{F} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational.

CLO2 U 4

4 a) Show that the vector field $\vec{v} = \frac{-x\hat{i} - y\hat{j}}{\sqrt{x^2 + y^2}}$ is a "sink field"

CLO3 App 5

b) Find the work done in moving a particle once around a circle C in xy plane, where the circle has center at the origin and radius 3 and the force field is given by $\vec{F} = (2x - y + z)\hat{i} + (x + y - z^2)\hat{j} + (3x - 2y + 4z)\hat{k}$

CLO3 App 5

5 a) Evaluate $\int_V \vec{F} dV$ where V is the region bounded by the planes:

CLO4 App 05

$x = 0, x = 2, y = 0, y = 3, z = 0, z = 4$ and $\vec{F} = xy\hat{i} + z\hat{j} - x^2\hat{k}$.

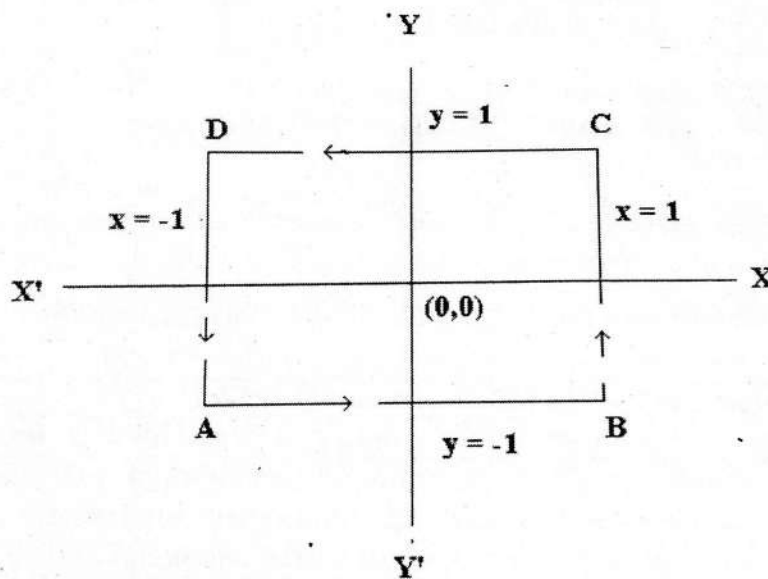
b) Determine the area enclosed by the graphs of $y = x^3$ and $y = 4x$ for $x \geq 0$

CLO4 App 05

Or

5

Verify Green's theorem for $\int_C (x^2 + xy)dx + (x^2 + y^2)dy$ where C is the square formed ABCD by the lines $y = \pm 1, x = \pm 1$ **CLO4 App 10**



International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc. in CSE Semester Final Examination, Spring 2026
Course Code: STAT 2311 Course Title: Probability and Statistics
Total marks: 50 Time: 2 hour 30 minutes

[Answer *all* the questions. Figures in the right-hand margin indicate full marks. Separate answer script must be used for Group A and Group B]

Group A

CO DL

1. a Explain the concept of correlation with a suitable example. What is the role of correlation in data analysis? 4 CO2 C2

OR

Define scatter diagram. Explain how it helps in studying correlation.

1. b A mobile app development team tracks weekly app crash reports and the number of bug fixes deployed: 6 CO2 C4

Week	1	2	3	4	5	6	7
Bug Fixes	4	6	3	5	7	6	2
Crash Reports	10	15	8	12	16	13	5

Analyze the relationship between the number of bug fixes and app crash reports by calculating the Pearson correlation coefficient. Interpret the strength and direction of the relationship, and discuss what the result implies about the software maintenance process.

OR

A data scientist records the number of training epochs (x) and the corresponding model accuracy (%) (y).

x	10	20	30	40	50
y	68	74	81	86	91

(i) Develop the linear regression equation for the given data and analyze the nature and strength of the relationship between training epochs and model accuracy; (ii) Predict the model accuracy after 60 training epochs.

2. a What is a random experiment? Discuss the importance of probability theory in computer science and engineering. 4 CO3 C2

2. b State Bayes theorem. A database system uses three query optimization algorithms, Q1, Q2, and Q3, to process SQL queries. They handle 45%, 20%, and 35% of all queries, respectively. The probabilities of producing an inefficient execution plan are 5%, 3%, and 4%, respectively. If a query receives an inefficient execution plan, (i) Find the probability that it was processed by Q2; (ii) Find the probability that it was processed by Q1 or Q3; (iii) Design a strategy for allocating future SQL queries among the three algorithms to minimize the overall probability of producing inefficient execution plans. Justify your proposed strategy using the computed probabilities. 6 CO3 C6

Group B

3. a Define a random variable. Distinguish between discrete and continuous random variables with suitable examples. 4 CO3 C2

OR

What is mathematical expectation? State four properties of expectation.

3. b The download time (in minutes) of a file follows the probability density function 6 CO3 C3

$$f(x) = Q(5 - x); \quad 1 \leq x \leq 5$$

(i) Determine the value of 'Q'; (ii) $P(X > 3)$; (iii) Find the expected download time (iv) $SD(5X - 6)$.

OR

Suppose that in a certain region of a country the daily rainfall (in inches) is a continuous random variable X with probability density function $f(x)$ given by

$$f(x) = P(6x - 3x^2); \quad 0 < x < 2$$

Compute (i) the value of 'P' (ii) $E[2X + 4.5]$ and (iii) $SD[X]$.

4. a Define the normal distribution. Explain the relationship between the Poisson and binomial distributions. 4 CO3 C2

4. b A website receives visitors randomly at an average rate of 5 visitors per minute. Assume the number of visitors follows a Poisson distribution. 6 CO3 C3
(i) Find the probability that exactly 7 visitors arrive in a minute.
(ii) Find the probability that no more than 4 visitors arrive in a minute.

5. a Discuss the concepts of alternative hypothesis, Type II error, level of significance and P value. 4 CO4 C2

5. b A regression model is expected to achieve an average accuracy of 90%. A random sample of 9 experimental runs gives a mean accuracy of 92.3% with a sample standard deviation of 2.4%. At the 5% significance level, test whether the average accuracy is greater than 90%. Note that the critical value of $t_{0.05,8} = 1.860$. 6 CO4 C5

International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc. in CSE Semester End Examination, Spring 2026
Course Code: CHEM-2301 Course Title: Chemistry
Total marks: 50 Time: 2 hour 30 minutes

[Answer *all* the questions. Figures in the right-hand margin indicate full marks. Separate answer script must be used for Group A and Group B]

Part A

- | | | | | |
|----|---|----|------|---|
| 1. | (a) Define an Electrochemical Cell. Describe the Daniel cell electrode potentials system with proper reaction. | C2 | CLO1 | 5 |
| | (b) Define Transference number and ionic mobility. The speed ratio of silver and nitrate ions in an aqueous solution of silver nitrate is 0.916. Calculate the transport number of both ions. | C4 | CLO2 | 5 |
| 2. | (a) What is a solution? Classify the different types of solutions. Discuss the components of a solution. | C2 | CLO1 | 5 |
| | (b) State and explain Henry's law with limitations. | C3 | CLO3 | 5 |

OR

- | | | | | |
|--|---|----|------|---|
| | (c) Define transport number and equivalence conductance. Show the variation of equivalence conductance with concentration of electrolytes in a solution. | C2 | CLO1 | 5 |
| | (d) State Nernst's distribution law. A solute distributes itself between water and benzene. At equilibrium, the concentration of the solute in water is 2 g/L, and in benzene it is 6 g/L. Find the distribution coefficient (K). | C3 | CLO3 | 5 |

Part B

- | | | | | |
|----|---|----|------|---|
| 3. | (a) What is a reversible reaction? Give one suitable example and "Chemical equilibrium is a dynamic process"- Explain. | C2 | CLO3 | 5 |
| | (b) State Le Chatelier's Principle. Explain how changes in concentration and temperature affect chemical equilibrium according to this principle. | C3 | CLO3 | 5 |
| 4. | (a) What is the order of a reaction? Explain zero-order, first-order, and second-order reactions with suitable examples. | C2 | CLO1 | 5 |
| | (b) How does temperature affect the rate of chemical reaction? Illustrate your explanation with a graph showing the relationship between temperature and reaction rate. | C4 | CLO3 | 5 |
| 5. | (a) What is adsorption and adsorption isotherm? Discuss about the types of adsorptions. | C2 | CLO1 | 5 |
| | (b) Derive Langmuir adsorption isotherm with assumptions. | C2 | CLO2 | 5 |

OR

- | | | | | |
|--|--|----|------|---|
| | (c) Define colloids and mention their classification based on physical state, and interaction with examples. | C2 | CLO1 | 5 |
| | (d) Mention the preparation methods of colloids. Explain any method in detail. | C2 | CLO2 | 5 |

International Islamic University Chittagong
Center for General Education (CGED)
Semester End Examination, Spring-26
Course Code: GEED-2302(GEED-2101 For LLB)
Course Title: Sciences of *Qur'an* and *Hadith*
Time: 2:30 Hours Full Marks: 50

Answer the following questions
(All questions are of equal value)

SL	Questions	Marks	CLOs	Bloom's taxonomy domain
1	Define <i>Makkai</i> and <i>Madani</i> revelations. Explain their characteristics along with the benefits of knowing them.	10	3	Remember & Create
2	Compare and contrast the various stages of <i>Jam' al - Qur'an</i> (preservation and compilation) during the periods of Prophet (SAAS), Abu Bakr (RA) and Uthman (RA) emphasizing their objectives and achievements .	10	3	Evaluate
3	Analyze the concept of <i>I'jaz al-Qura'n</i> . Critically evaluate the claim that scientific interpretations of the <i>Qur'an</i> provide evidence for the divine origin of the <i>Qur'an</i> .	10	3	Analyze & Evaluate
4	Define <i>Hadith</i> literally and terminologically. Explain the position and importance of <i>Hadith</i> in Islamic <i>Shari'ah</i> .	10	3	Remember & Create
5	a) Explain some types of <i>Hadith</i> according to the number of the reporters involved in each stage of <i>Sanad</i> with some examples. Or, b) Explain with examples: • <i>Sanad & Matan</i>, • <i>Al- Hadith As- Sahih</i>, 4 → • <i>Al- Hadith Al Mawdu'</i>, • <i>Al- Kutub As-Sittah</i> (six books of <i>Hadith</i>).	10	3	Create

International Islamic University Chittagong

Semester Ending Examination Spring: 2026

Course Title: Tazkiyatun Nafs

Course Code: MDP 2302A

Full Marks: 50 | Time: 2.5 Hours

PART: A | 20 Marks

Question No. 1. Choose the correct answer of the following:

10 Marks

- i. What is the linguistic meaning of the term Tazkiyah (تزكية)?
 - a) Purification and growth of the soul.
 - b) Political authority.
 - c) Material prosperity.
- ii. The ingredients of Tazkiyah are lying within the Kalimah Tayyebah
 - a) as it consists of believing in Allah.
 - b) as it consists of refusing all deities.
 - c) as it consists of negative and positive aspects of belief.
- iii. 'Ibadah in Islam :
 - a) is a concept of all encompassing concept.
 - b) is synonym to five pillars of Islam.
 - c) is synonym to loving and fearing Allah.
- iv. Dhikr, Du'a and recitation of the Qur'an is considered as:
 - a) 'Ibadah by tongue.
 - b) 'Ibadah by heart.
 - c) 'Ibadah by the organs.
- v. Salah, Sawm and Hajj are the:
 - (a) examples of 'Ibadah by the heart.
 - (b) examples of Ibadah by the wealth.
 - (c) examples of Ibadah the organs.
- vi. The primary purpose of mankind's creation is:
 - (a) scientific progress.
 - (b) economic development.
 - (c) worship of Allah alone.
- vii. The main purpose of Allah (SwT) behind the Khilafah is:
 - a) fighting in the way of Allah.
 - b) bestowing the mankind with special honor and superiority over the creation.
 - c) calling the people to the way of Allah.
- viii. As mentioned in the Quran Tazkiyah:
 - a) is the 1st objective behind sending the Prophet Muhammad (SAW).
 - b) is the 2nd and main objective.
 - c) the secondary objective.
- ix. The proper ways of attaining Tazkiyah is:
 - a) through freeing from the vices and adorning with the virtues.
 - b) through practicing regular Dhikr during the day and night.
 - c) through spending money to the poors and the distressed ones.
- x. The minor Shirk means:
 - a) Shirk in worship ('Ibadah).
 - b) believing that someone other than Allah has power like that of Allah.
 - c) Ria or Showing off.

Question No. 2. Write down the right statement of the following:

10 Marks

- i. **Tahliyah means:**
 - a) being free from vice of Shirk, Kafr and Nifaq
 - b) adorning the self with the qualities like God fearing, Tawakkul, Sabr and trustworthiness etc.
- ii. **The term good deeds includes:**
 - a) all obligatory acts and voluntary activities performed with sincerity.
 - b) honesty, justice, patience, and consistency.
- iii. **Trustworthiness requires:**
 - a) speaking the truth and being faithful to all.
 - b) adopting the quality of honesty, reliability, transparency, and respect for commitment and duties.
- iv. **Referring to knowledge and wisdom it can be said that:**
 - a) Knowledge is ability to know the people and wisdom is ability to lead them.
 - b) knowledge is acquisition of information and facts while wisdom is ability to apply that knowledge.
- v. **Strengthening the belief in Allah requires:**
 - a) to know about His existence.
 - b) to know about His Attributes.
- vi. **Among the requirements of reflecting an act after the deeds:**
 - a) whether the sincerity was exist or not.
 - b) whether the act was performed nicely or not.
- vii. **Muhasabat al-Nafs refers to :**
 - a) keeping the records of activities regularly.
 - b) practicing self accountability, self reflection and critical appreciation.
- viii. **Mujahadat al-Nafs refers to:**
 - a) internal spiritual struggle to purify ones inner self.
 - b) reflecting on his acts before and after the deeds.
- ix. **Nafs Ammrah is that soul:**
 - a) which commands the person to do evil deeds.
 - b) which reproaches the person on committing evil deeds.
- x. **Mujahadat an-Nafs can be attained:**
 - a) through regular self-reflection.
 - b) through leaving sinful activities and elements of attraction.

PART: B | 30 Marks

Answer any three of the following questions:

3x10 =30 Marks

- Q.1. What were the purposes of Allah behind creation of mankind? Explain.
- Q.2. Explain the concept of Tahliyah (تَحْلِيَة). Discuss the major principles upon which Tahliyah is founded and explain how these principles contribute to the moral and spiritual development of a Muslim.
- Q.3. What are the benefits of Muhasabat al-Nafs? Mention five of them elaborately.
- Q.4. What do you understand by the term Mujahadat al-Nafs? Explain with mentioning how it benefits you.
- Q.5. What do mean by the term "Exhortation"? Explain the Qur'anic approach of balancing between exhortation and intimidation with some examples.